

Superstore Sales & Profit Analysis

Business Intelligence Report

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Tools: Python (Pandas, Matplotlib, Seaborn), Kaggle, GitHub

Dataset: Sample Superstore (9,994 orders, U.S. retail)

Executive Summary

This report analyzes the Sample Superstore dataset across category, region, and customer segment, then drills into sub-category and discount behavior to identify where the business is winning, where it is leaking margin, and what to do about it.

Category Insights

- All three categories are profitable overall — none operates at a category-level net loss.
- **Technology** (~17.4% margin) and **Office Supplies** (~17.0% margin) are strong and comparable.
- **Furniture** generates sales on par with the other two categories (~\$742K) but converts to only **~2.5% margin** — a margin problem, not an outright loss.
- The margin gap concentrates in two sub-categories: **Tables** (net loss) and **Bookcases** (thin margin). The rest of Furniture (Chairs, Furnishings) is healthy.

Profitability Insights

- Overall profitability is **positive**: ~\$286K profit on ~\$2.30M sales (~12.5% blended margin).
- Discount level is **negatively correlated** with profit ($r = -0.22$); orders discounted above ~30% are where most loss-making transactions concentrate.

Regional Insights

- **West** leads in total sales (~\$725K), followed by East (~\$679K) and Central (~\$501K).
- **South** trails all regions in sales (~\$392K).
- **Central** underperforms on profit relative to its sales rank — a pattern worth investigating alongside the discount trend.

Segment Insights

- **Consumer** is the primary revenue driver, well ahead of Corporate and Home Office.
- Home Office is the smallest segment and a candidate for targeted promotions.

1. Category-Level Sales & Profit

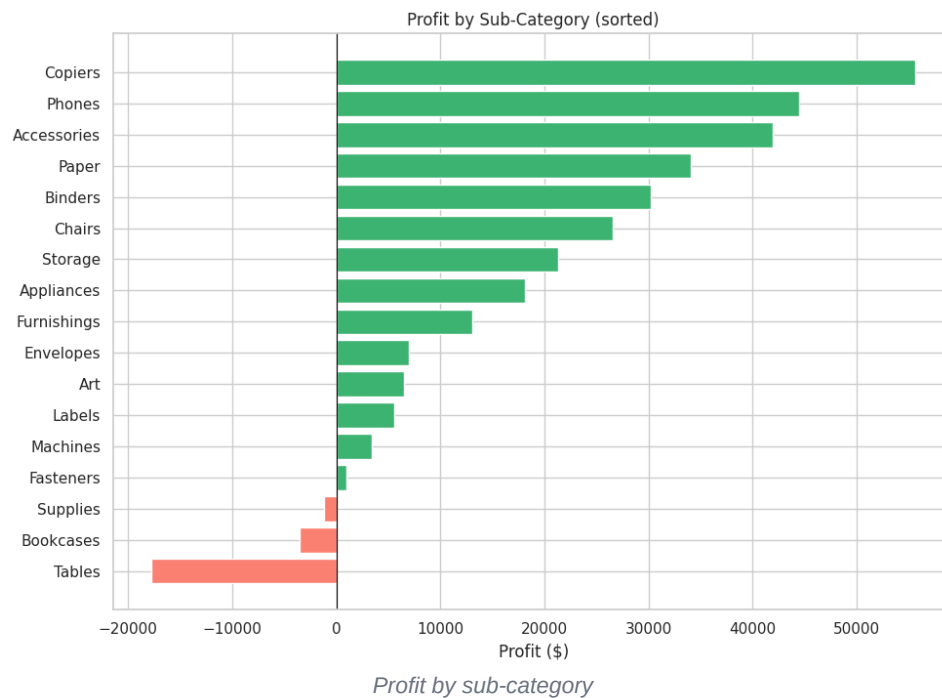
The question at this level isn't just which category sells the most — it's whether sales and profitability move together, since a top seller can still be a weak performer once margin is accounted for.



All three categories are profitable — none is a net loss at the category level. What differs sharply is margin: Technology and Office Supplies convert sales to profit at roughly the same efficient rate (~17%), while Furniture converts at only ~2.5% despite generating comparable sales. **Furniture isn't a demand problem, it's a margin problem** — and a margin problem is fixable with pricing/discount changes, a very different recommendation than dropping the category.

2. Sub-Category Breakdown

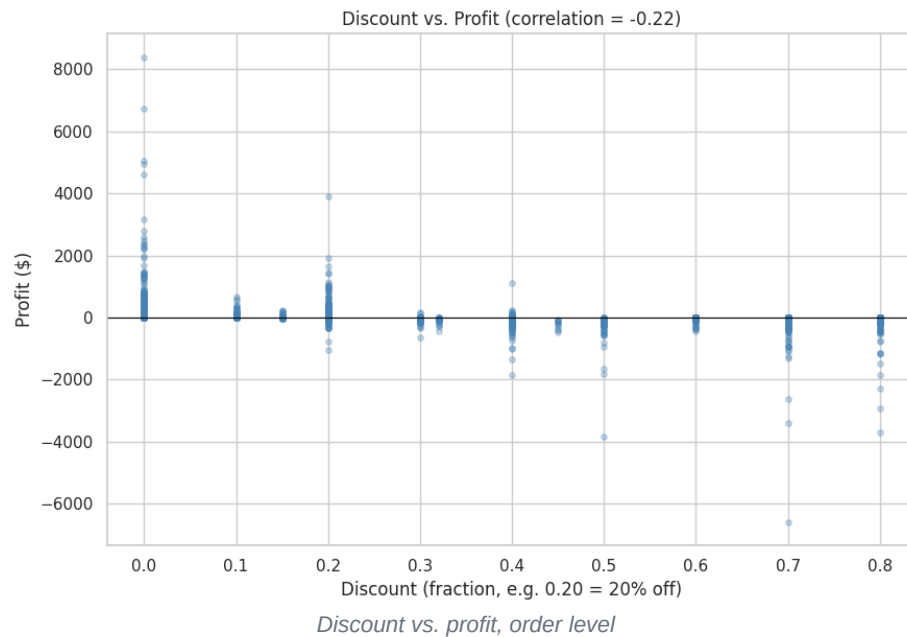
Category-level numbers hide where the problem specifically is. Dropping one level down turns “Furniture has a margin problem” into something a merchandising team can act on.



Only two sub-categories are actually unprofitable: **Tables** (a real net loss) and **Bookcases** (a small loss). Every other Furniture sub-category (Chairs, Furnishings) is solidly profitable — the issue is two identifiable sub-categories, not the category as a whole.

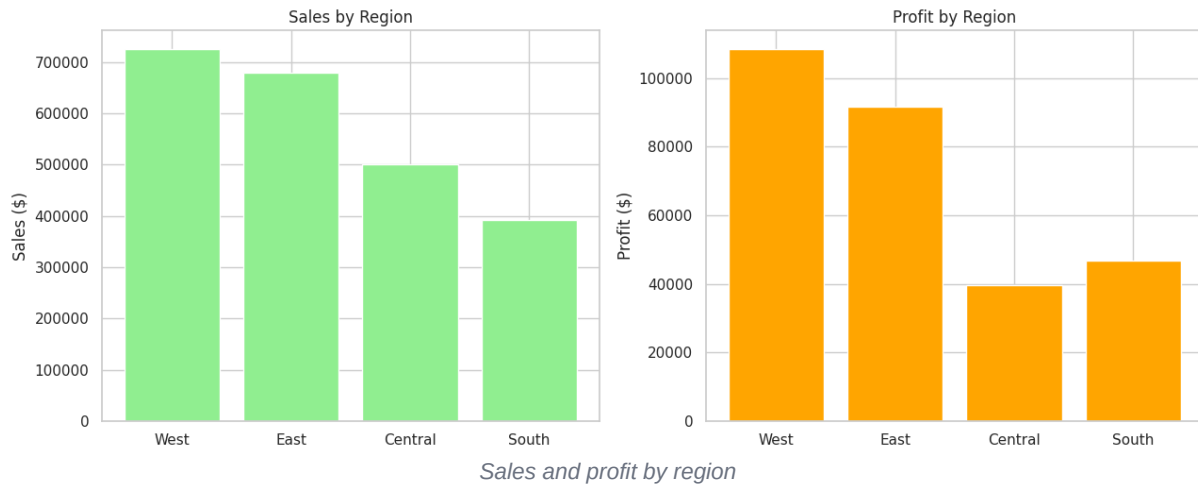
3. Discount vs. Profit Relationship

Retail intuition says heavy discounting erodes margin, but intuition isn't evidence. This section tests the relationship directly on order-level data.



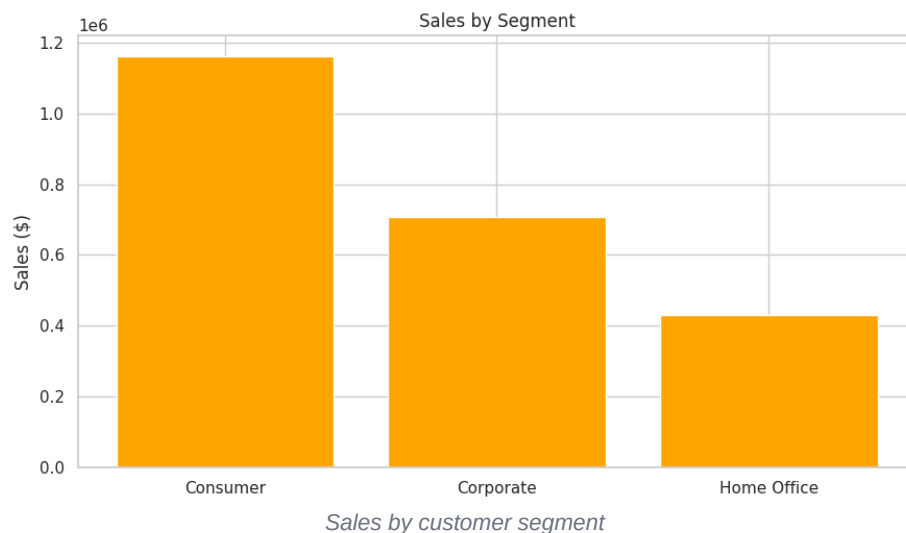
The correlation is moderate ($r = -0.22$) — discount alone doesn't fully explain profit, since product cost and category also matter. But the pattern is clear: **orders discounted above roughly 30% are where almost all negative-profit orders live.** Below that threshold, most orders stay profitable even with a discount applied — a specific, testable threshold a pricing team could act on.

4. Regional Performance



West leads in total sales (~\$725K), followed by East (~\$679K). **South** is the clear underperformer in sales (~\$392K), consistent across both sales and profit. One nuance: **Central has lower profit than South despite higher sales**, suggesting Central's issue is margin/discounting rather than demand — similar to the Furniture pattern above.

5. Segment Performance



Consumer is the primary revenue driver by a wide margin over Corporate and Home Office — the one finding from the original analysis that held up unchanged after re-checking the numbers.

6. Actionable Recommendations

1. Review pricing/discounting specifically for **Tables and Bookcases** rather than Furniture as a whole — that's where the real losses are.
2. Cap or restructure discounts above the **~30% threshold**, where profit turns negative for most orders.

3. Investigate why **Central** underperforms on profit despite mid-tier sales — likely a discounting pattern similar to Furniture's.
4. Investigate why **West** outperforms **East** in sales and apply learnings to under-indexed regions, especially **South**.
5. Continue leveraging Technology and Office Supplies' strong, comparable margins; evaluate whether Furniture pricing can be repositioned closer to their ~17% margin level.
6. Expand Home Office segment promotions to grow the smallest customer base.

Limitations

- This is a well-known public training dataset, not live business data — it doesn't carry the real-world messiness a production analysis would need to handle.
- Correlation between discount and profit is not causation; a controlled discount experiment (A/B test) would be needed to confirm the ~30% threshold as a real decision rule.